



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE
Batch : 2025-29

Assignment : 22

Subject: Programming with Java

ASSIGNMENT No : 22

TITLE : Demonstration and exploration of JDBC classes and interfaces (PreparedStatement, and ResultSet).

Problem Statement

Demonstration and exploration of JDBC classes and interfaces (PreparedStatement, and ResultSet).

Learning Objectives

To implement PreparedStatement and ResultSet for secure query execution and data retrieval.

Theory

PreparedStatement is an advanced JDBC interface used to execute precompiled SQL statements with parameters. Unlike the Statement interface, PreparedStatement allows SQL queries to be compiled only once and executed multiple times with different values, thereby improving performance. It also protects applications against SQL injection attacks by separating SQL code from user input. The ResultSet interface stores the records returned by SQL SELECT queries. It provides various methods such as next(), getInt(), getString(), and getDouble() for accessing retrieved data. PreparedStatement and ResultSet together provide secure, efficient, and flexible database interaction and are widely used in enterprise Java applications.

Code :

```
import java.sql.Connection;  
import java.sql.DriverManager;  
import java.sql.PreparedStatement;  
import java.sql.ResultSet;  
import java.util.Scanner;
```



```
public class Exp22 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Staff ID: ");
        String staffId = sc.nextLine();

        System.out.print("Enter Password: ");
        String password = sc.nextLine();

        String query = "SELECT role FROM hospital_staff WHERE staff_id = ? AND
password = ?";

        try (Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/hospital_db",
            "root",
            "Abcd@1234");
            PreparedStatement ps = con.prepareStatement(query)) {

            ps.setString(1, staffId);
            ps.setString(2, password);

            ResultSet rs = ps.executeQuery();

            if (rs.next()) {
                String role = rs.getString("role");
                System.out.println("Login Successful!");
                System.out.println("Access Granted: Welcome, " + role + " " + staffId);
            } else {
                System.out.println("Login Failed: Invalid Staff ID or Password.");
            }

        } catch (Exception e) {
            System.out.println("Database Error: " + e.getMessage());
        }

        sc.close();
    }
}
```



SQL Pre-Requisites :

```
1 • CREATE DATABASE hospital_db;
2 • USE hospital_db;
3 • CREATE TABLE hospital_staff (
4     id INT PRIMARY KEY AUTO_INCREMENT,
5     staff_id VARCHAR(50) UNIQUE NOT NULL,
6     password VARCHAR(50) NOT NULL,
7     role VARCHAR(20) NOT NULL -- 'Doctor' or 'Nurse'
8 );
9 • INSERT INTO hospital_staff (staff_id, password, role) VALUES
10 ('DOC205', 'medpass', 'Doctor'),
11 ('NUR310', 'nursepass', 'Nurse');
```

Output :

```
Problems 16 Output Debug Console Terminal Ports
• PS C:\Users\LENOVO\Desktop\JDBC\src> javac -cp "../lib/*" Exp22.java
• PS C:\Users\LENOVO\Desktop\JDBC\src> java -cp "../lib/*" Exp22
Enter Staff ID: NUR310
Enter Password: nursepass
Login Successful!
Access Granted: Welcome, Nurse NUR310
❖ PS C:\Users\LENOVO\Desktop\JDBC\src> |
```

Conclusion

The experiment successfully demonstrated JDBC-based database connectivity and staff authentication in Java. A MySQL database was connected using JDBC, and PreparedStatement was used to securely execute a parameterized SQL query. The program verified the entered Staff ID and password and displayed the appropriate login status and staff role. Thus, the experiment provided practical understanding of JDBC connectivity, SQL queries, PreparedStatement, ResultSet, and database-based authentication.



Exercise

Q1) Create a login application using PreparedStatement to check username and password.

Sol)

Code :

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;

public class Login {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Username: ");
        String username = sc.nextLine();

        System.out.print("Enter Password: ");
        String password = sc.nextLine();

        String query = "SELECT * FROM users WHERE username = ? AND password = ?";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/hospital_db",
                "root",
                "Abcd@1234"
            );

            PreparedStatement ps = con.prepareStatement(query);

            ps.setString(1, username);
            ps.setString(2, password);

            ResultSet rs = ps.executeQuery();
```



```
        if (rs.next()) {
            System.out.println("Login Successful!");
            System.out.println("Welcome, " + username);
        } else {
            System.out.println("Invalid Username or Password.");
        }

        rs.close();
        ps.close();
        con.close();

    } catch (Exception e) {
        System.out.println("Database Error: " + e.getMessage());
    }

    sc.close();
}
}
```

SQL Table :

```
27 • CREATE TABLE users (
28     id INT PRIMARY KEY AUTO_INCREMENT,
29     username VARCHAR(50) NOT NULL,
30     password VARCHAR(50) NOT NULL
31 );
32 • INSERT INTO users (username, password) VALUES ('doctor1', 'pass123');
```

Output :

```
Problems 20 Output Debug Console Terminal Ports
• PS C:\Users\LENOVO\Desktop\JDBC\src> javac -cp "../lib/*" Login.java
• PS C:\Users\LENOVO\Desktop\JDBC\src> java -cp "../lib/*" Login
Enter Username: doctor1
Enter Password: pass123
Login Successful!
Welcome, doctor1
❖ PS C:\Users\LENOVO\Desktop\JDBC\src> 
```



Q2) Create a Java application using PreparedStatement to authenticate hospital staff login (doctor/nurse/login ID and password). Display appropriate access messages based on authentication result.

Sol) Code :

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;

public class HospitalLogin {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Staff ID: ");
        String staffId = sc.nextLine();

        System.out.print("Enter Password: ");
        String password = sc.nextLine();

        String query = "SELECT role FROM hospital_staff WHERE staff_id = ? AND password = ?";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/hospital_db",
                "root",
                "Abcd@1234"
            );

            PreparedStatement ps = con.prepareStatement(query);

            ps.setString(1, staffId);
            ps.setString(2, password);

            ResultSet rs = ps.executeQuery();

            if (rs.next()) {
                String role = rs.getString("role");

                System.out.println("Login Successful!");

                if (role.equalsIgnoreCase("Doctor")) {
```



```
        System.out.println("Access Granted: Welcome Doctor " + staffId);
    } else if (role.equalsIgnoreCase("Nurse")) {
        System.out.println("Access Granted: Welcome Nurse " + staffId);
    } else {
        System.out.println("Access Granted: Welcome " + role + " " + staffId);
    }
} else {
    System.out.println("Login Failed: Invalid Staff ID or Password.");
}

rs.close();
ps.close();
con.close();

} catch (Exception e) {
    System.out.println("Database Error: " + e.getMessage());
}

}

sc.close();
}
}
```

SQL Table :

```
3 • CREATE TABLE hospital_staff (
4     id INT PRIMARY KEY AUTO_INCREMENT,
5     staff_id VARCHAR(50) UNIQUE NOT NULL,
6     password VARCHAR(50) NOT NULL,
7     role VARCHAR(20) NOT NULL -- 'Doctor' or 'Nurse'
8 );
9 • INSERT INTO hospital_staff (staff_id, password, role) VALUES
10 ('DOC205', 'medpass', 'Doctor'),
11 ('NUR310', 'nursepass', 'Nurse');
```



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Output : → Case 1 :

```
Problems 24 Output Debug Console Terminal Ports
• PS C:\Users\LENOVO\Desktop\JDBC\src> javac -cp "../lib/*" HospitalLogin.java
• PS C:\Users\LENOVO\Desktop\JDBC\src> java -cp ".;../lib/*" HospitalLogin
Enter Staff ID: DOC205
Enter Password: medpass
Login Successful!
Access Granted: Welcome Doctor DOC205
❖ PS C:\Users\LENOVO\Desktop\JDBC\src> 
```

→Case 2 :

```
Problems 24 Output Debug Console Terminal Ports
• PS C:\Users\LENOVO\Desktop\JDBC\src> javac -cp "../lib/*" HospitalLogin.java
• PS C:\Users\LENOVO\Desktop\JDBC\src> java -cp ".;../lib/*" HospitalLogin
Enter Staff ID: PUNE104
Enter Password: pune2007
Login Failed: Invalid Staff ID or Password.
❖ PS C:\Users\LENOVO\Desktop\JDBC\src> 
```